

Double-Reduction Helical Gearbox Teardown and Repair Analysis

A. Job information

Customer name

Work order number

Date received

Manufacturer

Model

Serial number

Overall reduction

: 1

Stage 1 ratio

: 1

Stage 2 ratio

: 1

Oil capacity

Oil grade

Application / driven equipment

Service hours (if known)

Reported failure / customer complaint

B. As-received condition

☐

Photos taken as received

Notes:

☐

External damage, cracks, missing parts

Notes:

☐

Nameplate present and legible

Notes:

Shafts turn freely by hand

☐

Yes

☐

No

Notes:

☐

Noise or roughness when turned

Notes:

☐

Oil leaks at seals / split line / plugs

Notes:

☐

Breather condition

Notes:

Ram Gear Gearbox Evaluation

C. Oil condition

Oil drained - volume recovered

Oil sample ID (lab analysis)

☐ Oil sample taken for lab analysis

Appearance: ☐ Clean ☐ Dark ☐ Milky (water) ☐ Sludge ☐ Metallic particles

Magnetic drain plug debris description

D. As-found measurements before teardown

Endplay - Input shaft

Endplay - Intermediate shaft

Endplay - Output shaft

Backlash - Stage 1

Backlash - Stage 2

Ext. runout - Input shaft

Ext. runout - Output shaft

E. Teardown checklist

- ☐ Match-mark housing, caps, and shafts before disassembly
- ☐ Parts tagged and kept in order
- ☐ Seals removed and inspected

AS-FOUND SHIM RECORD

Shaft	Drive end (DE) as-found shim (in/mm)	Non-drive end (NDE) as-found shim (in/mm)	Shim pack Replace / Reuse	
Input			☐ Replace	☐ Reuse
Intermediate			☐ Replace	☐ Reuse
Output			☐ Replace	☐ Reuse

Ram Gear Gearbox Evaluation

F. Component inventory and repair assessment

Input shaft

☐ Reuse ☐ Repair ☐ Replace

Part no.:

Findings:

Input pinion (stage 1)

☐ Reuse ☐ Repair ☐ Replace

Part no.:

Teeth:

Findings:

Intermediate shaft

☐ Reuse ☐ Repair ☐ Replace

Part no.:

Findings:

Intermediate gear (stage 1)

☐ Reuse ☐ Repair ☐ Replace

Part no.:

Teeth:

Findings:

Intermediate pinion (stage 2)

☐ Reuse ☐ Repair ☐ Replace

Part no.:

Teeth:

Findings:

Output shaft

☐ Reuse ☐ Repair ☐ Replace

Part no.:

Findings:

Output gear (stage 2)

☐ Reuse ☐ Repair ☐ Replace

Part no.:

Teeth:

Findings:

Bearing - Input DE

☐ Reuse ☐ Repair ☐ Replace

Cone P/N:

Cup P/N:

Findings:

Bearing - Input NDE

☐ Reuse ☐ Repair ☐ Replace

Cone P/N:

Cup P/N:

Findings:

Ram Gear Gearbox Evaluation

Bearing - Intermediate DE

☐

Reuse

☐

Repair

☐

Replace

Cone P/N:

Cup P/N:

Findings:

Bearing - Intermediate NDE

☐

Reuse

☐

Repair

☐

Replace

Cone P/N:

Cup P/N:

Findings:

Bearing - Output DE

☐

Reuse

☐

Repair

☐

Replace

Cone P/N:

Cup P/N:

Findings:

Bearing - Output NDE

☐

Reuse

☐

Repair

☐

Replace

Cone P/N:

Cup P/N:

Findings:

Input seal

☐

Reuse

☐

Repair

☐

Replace

Part no.:

Findings:

Output seal

☐

Reuse

☐

Repair

☐

Replace

Part no.:

Findings:

Keys and keyways

☐

Reuse

☐

Repair

☐

Replace

Part no.:

Findings:

Spacers and locknuts

☐

Reuse

☐

Repair

☐

Replace

Part no.:

Findings:

Bearing caps and shims

☐

Reuse

☐

Repair

☐

Replace

Part no.:

Findings:

Ram Gear Gearbox Evaluation

Housing and bearing bores

☐

Reuse

☐

Repair

☐

Replace

Part no.:

Findings:

Breather, plugs, sight glass

☐

Reuse

☐

Repair

☐

Replace

Part no.:

Findings:

Other:

☐

Reuse

☐

Repair

☐

Replace

Part no.:

Findings:

Other:

☐

Reuse

☐

Repair

☐

Replace

Part no.:

Findings:

Other:

☐

Reuse

☐

Repair

☐

Replace

Part no.:

Findings:

G. Failure modes observed

Gears

☐

Pitting

☐

Spalling

☐

Scuffing/scoring

☐

Wear

☐

Tooth breakage

☐

Plastic deformation

☐

Corrosion

☐

Misaligned contact pattern

Bearings

☐

Spalling/fatigue

☐

Brinelling

☐

Contamination dents

☐

Overheating/discoloration

☐

Corrosion/etching

☐

Electrical fluting

☐

Cage damage

☐

Skidding

Location / notes

Ram Gear Gearbox Evaluation

H. Dimensional checks

BEARING BORE AND SHAFT JOURNAL DIAMETERS

Shaft	Loc.	Bearing bore measured	Bearing bore spec	Bore OK	Shaft journal measured	Shaft journal spec	Journal OK
Input	DE			☐			☐
Input	NDE			☐			☐
Intermediate	DE			☐			☐
Intermediate	NDE			☐			☐
Output	DE			☐			☐
Output	NDE			☐			☐

Seal journal condition notes

I. Root cause

- ☐ Lubrication failure ☐ Contamination ☐ Overload / shock load ☐ Misalignment
- ☐ Improper installation ☐ Fatigue / end of life ☐ Manufacturing defect
- ☐ Other

Root cause narrative

J. Repair summary

Parts to replace

Machining / repair work required

Recommended upgrades or changes

Estimated labor hours

Estimated parts cost

Quote number

Recommendation:

☐

Repair

☐

Replace unit

☐

Not economical to repair

K. Sign-off

Inspected by - name

Signature

Date

Reviewed / approved by - name

Signature

Date

☐

Customer approval received

Date: